HAIRUL AZHAR BIN ABU BAKAR

FORWARD-DEPLOYED AI ENGINEER | AI SOLUTIONS ARCHITECT | FOUNDER

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PROFILE: Forward-deployed AI engineer and founder who builds agentic AI and runs it in production. Through Agentic Intelligence Matrix (AiM Intelligence), Singapore, I architect **KatalockAi**, a live multilingual AI workspace for India and Southeast Asia; its **523-lesson Academy**; and **AIM OS**, a governed agent platform. I have shipped live client AI products on KatalockAi. Before AI, I spent 10+ years owning budgets, SLAs and on-site delivery in integrated-resort fit-outs, luxury-hospitality supply and enterprise logistics, so I build AI that operators can trust: verified, human-approved and recoverable.

LIVE PRODUCTS & CLIENT DELIVERY

KatalockAi
www.katalock.net
Agentic workspace that researches, builds, markets and creates media in local languages across India and SEA, with K-Computer verification.

KatalockAi Academy
www.katalock.net/academy
20 phases, 523 lessons, 6 career routes; Professor Katalock tutors, reviews and grades student code.

AIM OS
aimglobal.io
Governed agent OS: 50+ apps, 108 API endpoints, 27+ agents, QGov policy engine and killswitch.

IBTISAM · Katalock for Education
www.katalock.net/edu
AI tutor for schools: explains, draws and quizzes, never does the homework; teachers see mastery per student.

Tarot with Shilpa · client AI product
www.tarotwithshilpa.com
Persona-based AI readings in the user's language, freemium metering and hand-off to a human reader.

Ignis Eco · client website
www.igniseco.com
Investor site for a plastic-to-fuel venture: process, plant and team, with a language switcher.

TECHNICAL SKILLS & ARCHITECTURE STACK

AI & Agentic Engineering

Agentic AI · Multi-Agent Orchestration · Cross-Model Verification · Mixture-of-Agents Orchestration · RAG · Model Context Protocol (MCP) · Knowledge Distillation · Quantization · Evaluation & Release Gates · Human-in-the-Loop Approval · AI Safety & Red-Teaming · AI Tutoring Guardrails · Voice AI & Telephony (STT, TTS, Twilio) · Vector DBs (Pinecone, Chroma)

Full-Stack & Systems Infrastructure

Python · TypeScript · React / Next.js · Node.js / Express · FastAPI / Django · PostgreSQL / Supabase · Redis · Docker / Kubernetes · AWS / GCP · Terraform · Stripe

PROFESSIONAL EXPERIENCE

AI Platform Architect & Co-Founder | KatalockAi (www.katalock.net) — **2026 – Present**
Agentic Intelligence Matrix (AiM Intelligence) · Singapore / India · Remote

- Own backend architecture, agent orchestration and infrastructure end to end for an agentic AI workspace built by a small team across Singapore and India; co-founder runs on-the-ground operations.
- Built **K-Computer**, the orchestration layer under Workspace, Code and Studio: a live cross-model verification panel where models from different labs answer independently and separate judges reconcile one answer with an agreement score; it also reviews apps before they ship.
- Designing its next stage: a four-tier Mixture-of-Agents pyramid (Lucy, Tara, Stargazer, AiM) with per-tier arbiters, consensus-gated escalation and a distillation flywheel; design targets ~85% of tasks at the base, ~10x lower cost.
- Shipped an OpenAI-compatible API with named tiers and per-call cost, GST/UPI and currency localisation for three markets, payment-gated browser automation and a code sandbox that pushes to GitHub.
- Built **Katalock Live**, spoken conversation with the workspace: replies stream out loud sentence by sentence after 0.7 s of quiet, the user can interrupt by talking, hands-free commands work in 8 languages, and tables and diagrams are talked through rather than read out; ten voices with selectable personality.

- Built **voice agents** that answer a business's own phone number and WhatsApp: live interruptible calls streamed through the build server (silence detection, Whisper transcription, sentence-by-sentence speech), answers from a daily-refreshed website fact sheet, Google Calendar booking without double-booking, complaint logging, colleague checks and warm transfer to a human, billed per second.
- Human approval by design: no message, payment or post leaves without a person saying yes. Priced from ₹299/month to done-for-you agent teams.
- Delivered client AI products: **Tarot with Shilpa** (persona-based readings, freemium metering, human hand-off), the **Ignis Eco** investor site, AI video for property marketing, and SME bundles for clinics, retail, real estate and restaurants.

Architect & Founder | KatalockAi Academy & Katalock for Education **2026 – Present**
www.katalock.net/academy · www.katalock.net/edu · Remote

- Built a **20-phase, 523-lesson** AI engineering platform with 6 career routes, 6 skill paths and 17 certified capstones; exams issue certificates with verifiable IDs.
- Built **Professor Katalock**, a Socratic AI tutor in every lesson that checks understanding, reviews student code, flags edge cases and grades projects against rubrics.
- Curriculum spans maths to production agents: LLM serving (vLLM, SGLang, TensorRT-LLM), quantization (AWQ, GPTQ, FP8), alignment (RLHF, DPO, Constitutional AI), MCP and multi-agent systems.
- Built **IBTISAM** (Katalock for Education), an AI tutor for schools that teaches without doing the homework: teacher-set help levels, manipulation-resistant refusals, quizzes scored per objective, rubric marking drafts and exam mode.

Founder & AI Platform Architect | Agentic Intelligence Matrix · AIM OS (aimglobal.io) **Jan 2023 – Present**
Singapore · Remote

- Founded a Singapore AI company rooted in **AI safety and red-teaming**: stress-testing models to find where they hallucinate, leak and fail, then building guardrails first.
- Architected AIM OS, an operating system for agentic AI spanning 50+ integrated apps, 108 API endpoints, 27+ autonomous agents and about 80 database tables.
- Designed the **QGov** policy engine, **Sentinel** safety monitor and a founder killswitch; built **Nexus Hub** orchestration with intent routing, pg-boss retries and Yjs CRDT collaboration.

Co-Founder, Investor Relations & GTM | Ignis Eco (www.igniseco.com) **2026 – Present**
Tamil Nadu / Bengaluru, India

- Lead investor relations, seed-stage capital structuring and go-to-market for a plastic-to-fuel pyrolysis venture.

Logistics Technology Operations (contract) | GOGOVAN Singapore (GOGOX) **Jul 2024 – Sep 2024**
Singapore

- Ran enterprise logistics technology for IKEA Singapore: **99.8% on-time delivery** across 5,000+ daily orders, with incident resolution 25% faster.

Co-Founder & Operations Director | Freda'D Pte Ltd / DGCC Investment Pte Ltd **Apr 2014 – Dec 2024**
Singapore / Middle East · run concurrently with employment roles

- Grew a bootstrapped fragrance business from $500 to a profitable international supplier to luxury hotel portfolios in Asia and the Middle East, at **25% YoY growth**.
- Deployed automation and inventory systems in client operations, cutting overhead 30% and lifting engagement 40% at 99.8% on-time delivery.

Project Director | HH Design Pte Ltd / Interior Producer Pte Ltd **Jan 2010 – Mar 2020**
Singapore

- Delivered integrated-resort fit-outs including **Marina Bay Sands Casino** and **Resorts World Sentosa**, owning $500K–$5M budgets and improving timelines by 15%.

FEATURED PROJECTS & SYSTEMS

1. KatalockAi Workspace www.katalock.net
Next.js, TypeScript, Supabase/PostgreSQL, Composio, Docker, Stripe
Multilingual agentic workspace for India and Southeast Asia: chat, deep research, app building with a security check, Studio workflows, agent crews and media, priced in INR, SGD, MYR and IDR.

2. K-Computer Workspace · Code · Studio
Multi-model panel, source checks, judge reconciliation
Cross-model verification live in chat and Code, returning one answer with an agreement score. Next: a four-tier Mixture-of-Agents pyramid (design targets ~85% at the base, ~10x lower cost).

3. Katalock Live & Voice Agents www.katalock.net/voice
Twilio, WebSockets, Whisper, streaming TTS, Google Calendar
Interruptible spoken conversation with the workspace, plus AI receptionists on a business's own phone and WhatsApp that book, log complaints, check with colleagues and warm-transfer.

4. KatalockAi Academy www.katalock.net/academy
Next.js, Supabase, automated evaluation, rubric grading
20-phase, 523-lesson AI engineering platform where Professor Katalock reviews code and grades projects; certificates with verifiable IDs.

5. IBTISAM · Katalock for Education www.katalock.net/edu
Tutor guardrails, objective-level mastery, exam mode
AI tutor for schools that teaches without doing the homework, with teacher-set help levels, quizzes scored per objective and rubric marking drafts, in 11 languages.

6. AIM OS aimglobal.io
Python, Node.js, PostgreSQL, Redis, Docker, Kubernetes, Yjs
Governed agent operating system: 50+ apps, 108 API endpoints, 27+ agents, with the QGov policy engine, Sentinel safety monitor and Nexus Hub orchestration.

7. Tarot with Shilpa www.tarotwithshilpa.com
Built with KatalockAi: LLM personas, multilingual UI, auth, metering
Consumer AI reading app with reader personas, spreads, a monthly free tier and escalation to a human reader.

8. WorkflowGenius
github.com/CybergigSG/WorkflowGenius
Python, RAG, Pinecone, React, AWS Lambda/EC2
Enterprise RAG pipeline processing 10,000+ daily transactions at 99.5% uptime, cutting admin workload 40%.

EDUCATION & CREDENTIALS

- **MSc in Computer Science**, Heriot-Watt University (expected May 2028)
- **Diploma in Architectural Technology**, Singapore Polytechnic (2001–2004)

PROFESSIONAL CERTIFICATIONS

- **IBM Full Stack Software Developer** (2024): React, Node.js, Python, Django, Docker, Kubernetes, OpenShift
- **Google AI & Machine Learning** (2024): generative AI for software development, Python data science
- **Google Cybersecurity Professional Certificate** (2024)

ADDITIONAL INFORMATION

Work authorisation: Singapore citizen. Open to relocation: USA, UK, EU, UAE, Australia, Canada, India, APAC.
Languages: English (native), Malay (native) · **Availability:** immediate, or two weeks' notice